Multimedia Signal Processing - 2nd Module

January 16, 2025

1. **Question (7 pts)**
   Consider a continuous source X that emits symbols following a probability distribution function (p.d.f.) $p_X(x)$.
   (a) Consider the following series of symbols emitted by X

   $$x = \{-1.6,\ 0.2,\ -1.6,\ -1.4,\ 1.4,\ -1.4,\ -0.2,\ 0.4,\ 1.6,\ -0.4,\ 2.5\}.$$

   Design a Lloyd-Max quantizer tailored to these symbols using $M = 3$ bins and the thresholds $\{-\infty, -1.0, 0.6, +\infty\}$ as initialization. Stop after one complete iteration. Show all the passages.
   (b) Quantize x using the Lloyd-Max quantizer designed at point (a), thus obtaining $\tilde{x}$.
   (c) Compute the entropy of the source representing $\tilde{x}$ estimating the probability mass function (p.m.f.) directly from $\tilde{x}$.
   (d) Considering the source that generated $\tilde{x}$, propose a one-to-one code that is not uniquely decodable.

2. **Question (5 pts)** Consider the problem of image coding.
   (a) With the help of a block diagram, describe the JPEG image encoding scheme (*note: no need to describe the decoder*).
   (b) Comment the diagram explaining which is the input and the output of each block.
   (c) Comment on which blocks take care of visual irrelevancy, and which blocks take care of visual redundancy.

3. **Question (3 pts)** Consider two source X and Y characterized by a finite element alphabet.
   (a) Explain without using any equation the meaning of the entropy of X.
   (b) Show with a Venn diagram the relationship between the following quantities: $H(X)$, $H(Y)$, $H(X|Y)$, $H(X,Y)$, $I(X,Y)$.

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Consider a video coding scheme, motion vectors are always the same despite the algorithm used to estimate them.
   - [T] [F] Given a transform coding scheme, DCT is typically faster to be computed than KLT.
   - [T] [F] The absolute threshold of hearing can be used for bit allocation in audio coding.
   - [T] [F] A scalar uniform quantizer is an optimal choice for uniformly distributed sources.
   - [T] [F] Considering a transform coding scheme, it always provide better rate-distortion results of a PCM scheme.
   - [T] [F] There exists a single unique Huffman Code for each finite alphabet source.
   - [T] [F] When using linear predictive vocoder exploiting the source-system model, the coefficients of the synthesis filter are directly quantized.
   - [T] [F] Lossless coding techniques cannot be applied to sources characterized by an infinite alphabet.

5. **Question (Matlab - 10pts)**

   Consider you have an image file called `peppers.png` in your Matlab working directory. This is a RGB image with resolution 512×384.

   Data loading

   (a) Load the image and display it.

   (b) Crop the top-left 128×128 pixels of the image (use the cropped version of the image hereinafter).

   (c) Convert to grayscale and display the image again. Feel free to chose the RGB-to-gray conversion you want.

   DPCM-based coding

   Considering the color component you extracted from the image, implement a closed loop DPCM coding scheme.

   (d) Turn the grayscale cropped image into a monodimensional vector.

   (e) Considering that the prediction of the current sample is the previous sample, define a uniform scalar quantizer that works with (approximately) 16 levels (*hint: use an open-loop estimate of the signal you are going to quantize*).

   (f) Encode the signal with the DPCM.

   (g) Decode the signal.

   (h) Compute the PSNR.